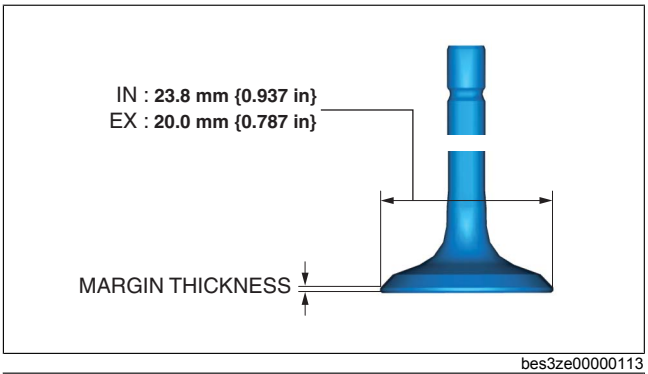
VALVE, VALVE GUIDE INSPECTION

id011000506200

1. Measure the valve head margin thickness using a vernier caliper.
   - If it is less than the standard specification, replace the valve.

   Standard valve head margin thickness
   IN: 1.50—1.70 mm {0.0591—0.0669 in}
   EX: 1.68—1.88 mm {0.0662—0.0740 in}

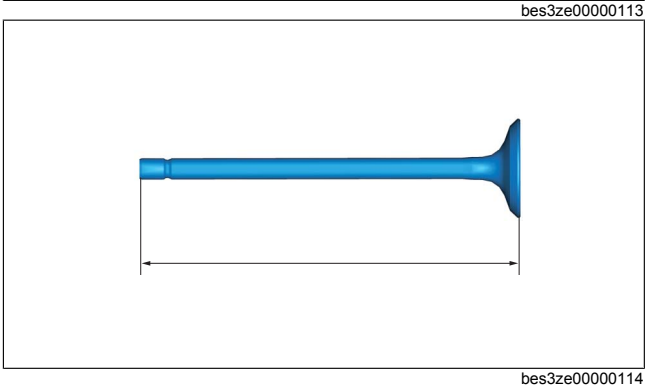


bes3ze00000113

2. Measure the length of the each valve using a vernier caliper.
   - If it is less than the minimum specification, replace the valve.

   Standard valve length
   IN: 97.07—97.37 mm {3.822—3.833 in}
   EX: 96.99—97.29 mm {3.819—3.830 in}

   Minimum valve length
   IN: 96.85 mm {3.813 in}
   EX: 96.77 mm {3.810 in}

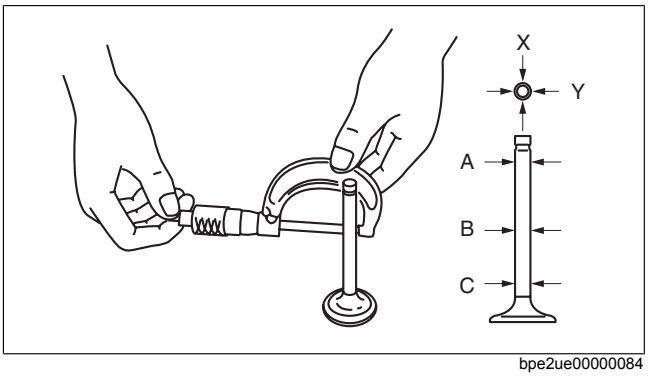


bes3ze00000114

3. Measure the valve stem diameter of each valve using the micrometer. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.
   - If it is less than the minimum specification, replace the valve.

   Standard valve stem diameter
   IN: 5.470—5.485 mm {0.2154—0.2159 in}
   EX: 5.465—5.480 mm {0.2152—0.2157 in}

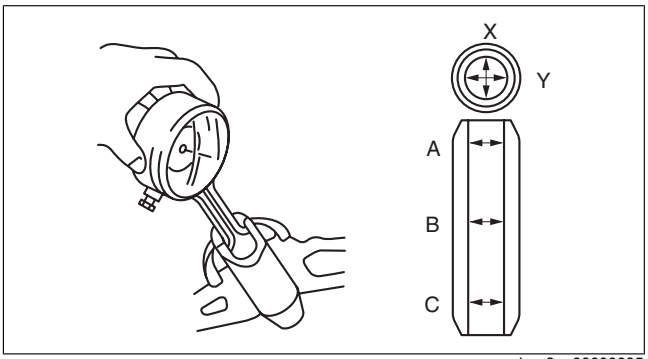
   Minimum valve stem diameter
   IN: 5.424 mm {0.2135 in}
   EX: 5.419 mm {0.2133 in}



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4. Measure the inner diameter of each valve guide using the caliper gauge. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.
   - If it is not within the specification, replace the valve guide.

   Standard valve guide inner diameter
   IN: 5.510—5.530 mm {0.2170—0.2177 in}
   EX: 5.510—5.530 mm {0.2170—0.2177 in}

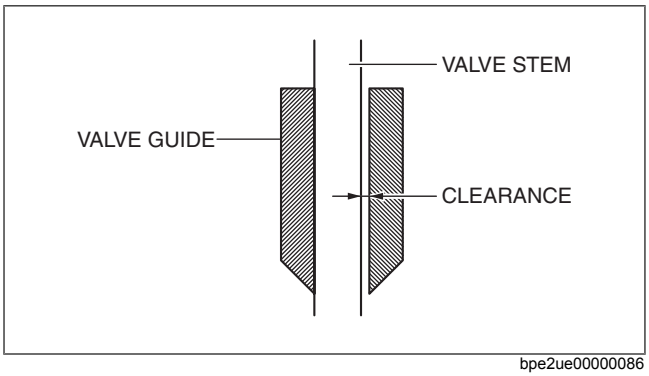


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5. Calculate the clearance between the valve stem and the valve guide by subtracting the inner diameter of the valve guide from the outer diameter of the corresponding valve stem.
   - If it exceeds the maximum specification, replace the valve or valve guide.

   Standard clearance between valve stem and guide
   IN: 0.025—0.060 mm {0.0010—0.0023 in}
   EX: 0.030—0.065 mm {0.0012—0.0025 in}

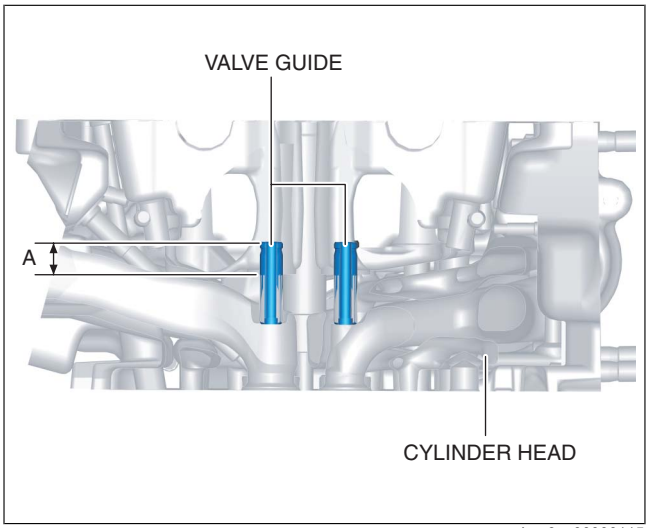
   Maximum clearance between valve stem and guide
   0.10 mm {0.0039 in}



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6. Measure the projection height (dimension A) of each valve guide using the vernier caliper.
   - If it is not within the specification, replace the valve guide.

   Standard valve guide projection height
   14.8—15.3 mm {0.583—0.602 in}



bes3ze00000115